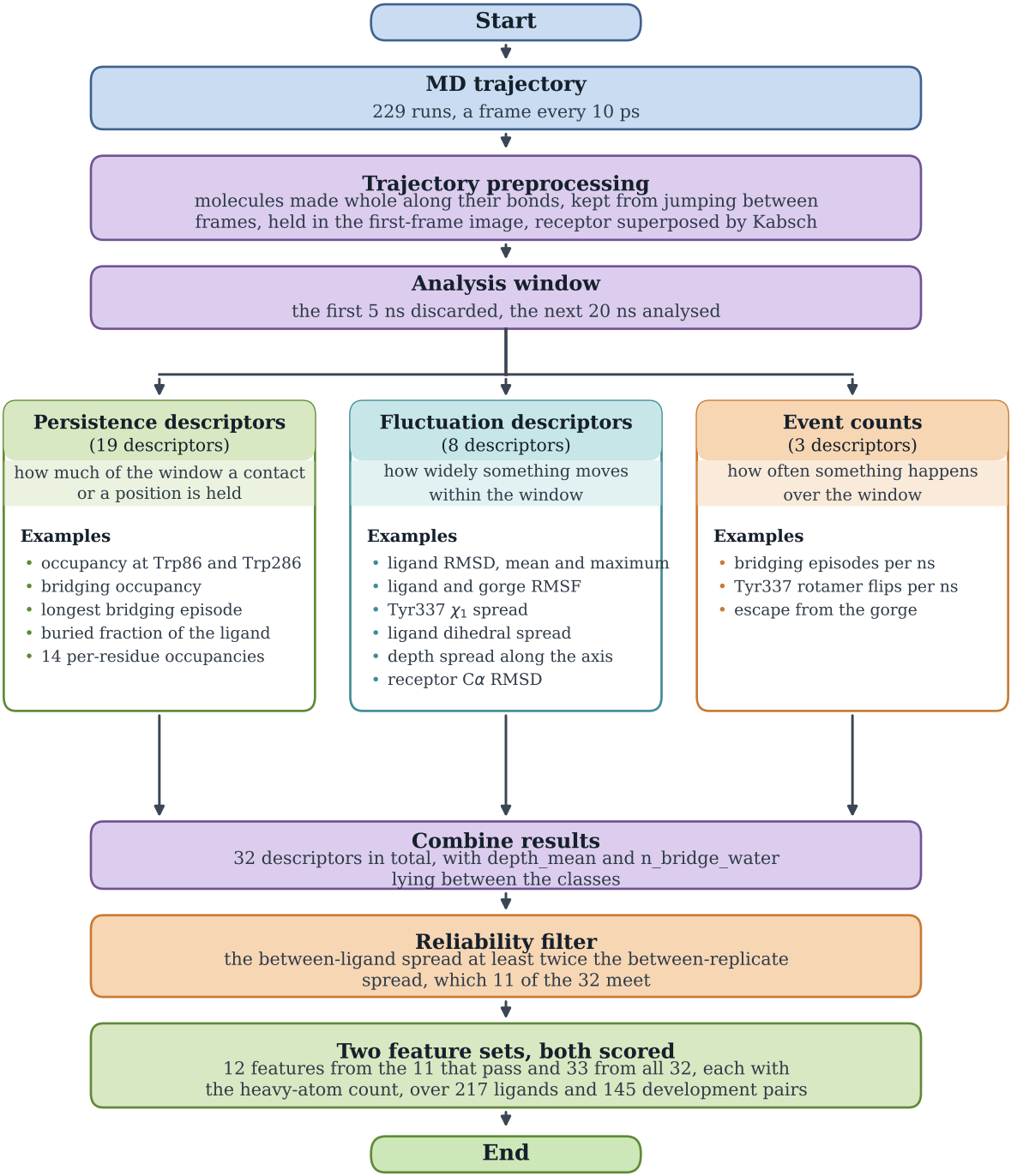
Start
MD trajectory
229 runs, a frame every 10 ps
Trajectory preprocessing
molecules made whole along their bonds, kept from jumping between frames, held in the first-frame image, receptor superposed by Kabsch
Analysis window
the first 5 ns discarded, the next 20 ns analysed
Persistence descriptors
(19 descriptors)
how much of the window a contact or a position is held
Examples
occupancy at Trp86 and Trp286
bridging occupancy
longest bridging episode
buried fraction of the ligand
14 per-residue occupancies
Fluctuation descriptors
(8 descriptors)
how widely something moves within the window
Examples
ligand RMSD, mean and maximum
ligand and gorge RMSF
Tyr337 χ1 spread
ligand dihedral spread
depth spread along the axis
receptor Cα RMSD
Event counts
(3 descriptors)
how often something happens over the window
Examples
bridging episodes per ns
Tyr337 rotamer flips per ns
escape from the gorge
Combine results
32 descriptors in total, with depth_mean and n_bridge_water lying between the classes
Reliability filter
the between-ligand spread at least twice the between-replicate spread, which 11 of the 32 meet
Two feature sets, both scored
12 features from the 11 that pass and 33 from all 32, each with the heavy-atom count, over 217 ligands and 145 development pairs
End